

Project Report
On

FINDLY - CAMPUS LOST & FOUND



U.V. Patel
College of
Engineering

**CENTRE OF
EXCELLENCE**
NOTIFIED
BY GOVERNMENT OF GUJARAT



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ABSTRACT

Traditional lost-and-found handling on most college campuses still relies on notice boards, verbal word-of-mouth, and informal WhatsApp groups; postings are not searchable, there is no permanent record of what was lost or found, and - most importantly - there is no way to verify that a person claiming an item is actually its rightful owner, so belongings are frequently handed to the wrong person or simply never recovered. **FINDLY - Campus Lost & Found** overcomes these limitations with a centralised, web-based platform that gives every reported item a searchable digital record and routes every found item through a verified custody chain: found items can be logged into the system only by Staff or Admin accounts, mirroring how a physical item is actually handed in at a security desk or admin office, while students can report items they have lost and browse items that have been found. When the rightful owner comes forward, Staff/Admin verify the person face-to-face and record their name, contact number and enrollment number before the item is handed over - no separate claim-submission step is required. The system follows a standard three-tier web architecture, with a front-end built using **HTML5, CSS3, JavaScript** and **Bootstrap**, and a back-end built using **PHP** with a **MySQL** database, and is organised into four **core modules**: Authentication & Account Management, Lost-Item Posting (Student), Found-Item Logging & Custody Tracking (Staff/Admin only), and Item Resolution & Handover (Staff/Admin recording verified recipient details).

Beyond routine account access, the platform's major features include role-restricted posting of found items so that only Staff/Admin custodians can log what has physically been handed in, a full custody-tracking trail that records where a found item is currently held until it is resolved, an in-person recipient-verification step that captures the claimant's name, contact number and enrollment number at the point of handover, multi-filter category/date/location search, real-time in-app status notifications for posters, and an Admin dashboard summarising pending approvals, pending handovers, resolved items and registered users, backed by a granular role-permission matrix that governs exactly which functions each of Student, Staff and Admin may perform. By replacing an untracked, informal manual process with a verifiable, searchable and role-based digital workflow, FINDLY is expected to shorten the time it takes to reunite students with lost belongings, and give campus administration complete visibility and accountability over the entire lost-and-found process.

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1. INTRODUCTION

1.1 OVERVIEW

Every academic institution, regardless of size, deals with a continuous stream of misplaced belongings such as ID cards, wallets, mobile phones, bottles, textbooks, bags, and electronic accessories. These items may be lost in classrooms, libraries, canteens, hostels, laboratories, sports areas, or other locations within the campus. In many institutions, the process of reporting and finding such belongings is still handled through informal methods such as asking other students, checking physical notice boards, contacting staff members, or sharing information through messaging groups. These approaches can make it difficult to maintain a central record of lost and found items and may require users to check multiple sources before finding relevant information. **FINDLY - Campus Lost & Found** is proposed as a centralized, web-based platform for managing lost and found items within an academic campus.

FINDLY provides a common platform through which students can report items they have lost and browse items that have been found on campus. Found-item records are handled by authorized **Staff and Admin users**, who are responsible for recording found belongings in the system. This ensures that the process of registering a found item is associated with an authorized campus role rather than allowing every user to create a found-item record. Students can view available found-item information and contact the appropriate campus authority for further handling of their belongings. The system follows **role-based access control**, where different users are provided with different modules and operations according to their role. This helps maintain a clear separation between student activities and staff or administrative activities.

The system is designed as a web-based and responsive application so that users can access it through commonly available devices such as computers, laptops, tablets, and smartphones. Instead of depending on physical registers, notice boards, or separate messaging groups, FINDLY provides a structured platform where lost and found records can be stored and searched in one place. Users can enter relevant information such as item name, category, description, location, date, and image while creating or managing an item record. Search and filtering functionality allows users to find relevant found-item records more conveniently. The system is intended to simplify the process of reporting, searching, recording, and managing campus lost-and-found information.

The system mainly consists of three user roles: **Student, Staff, and Admin**. Students form the primary user group for reporting lost belongings and searching for found items. Their module includes the student dashboard, reporting lost items, viewing and managing their own lost reports, browsing found items, searching and filtering found items, viewing item details, notifications, and profile management. Staff members are responsible for handling found belongings that are handed over to the institution. Their module includes logging found items, managing found-item records, updating the custody location of an item, marking an item as claimed or resolved, and recording the recipient's relevant details such as name, contact number, and enrollment or employee number. Admin users have administrative control over the system and can perform staff-level operations along with functions such as managing users, managing categories, managing lost and found records, and viewing system-level information and statistics.

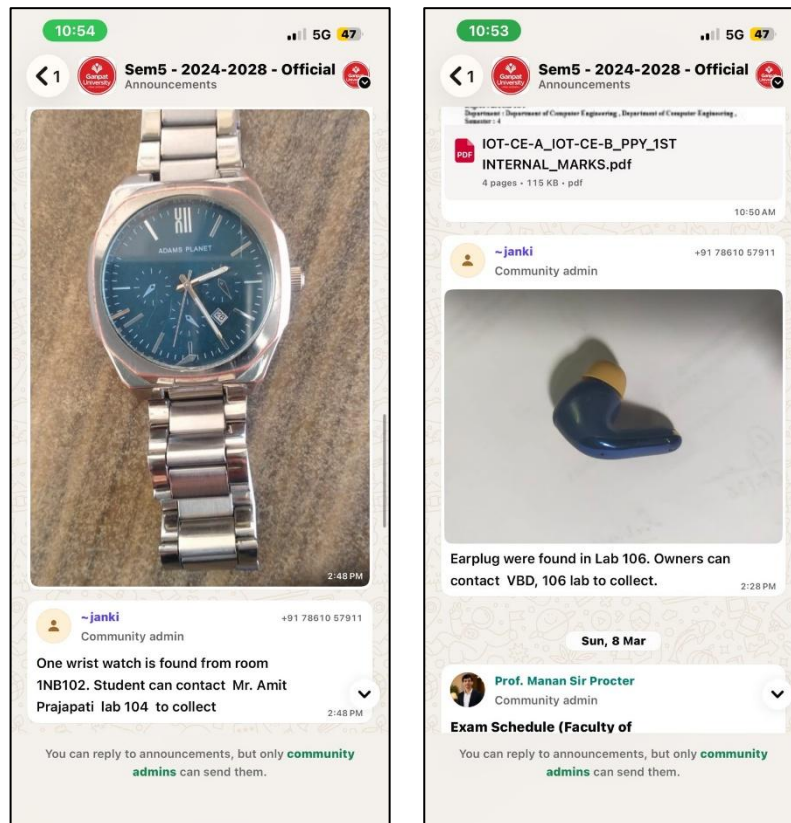
FINDLY therefore provides a structured approach to campus lost-and-found management by bringing the reporting, searching, recording, and management of belongings into a single platform. The proposed system aims to reduce dependence on informal communication methods and provide a clearer workflow for students, staff, and administrators while maintaining appropriate access control for each role.

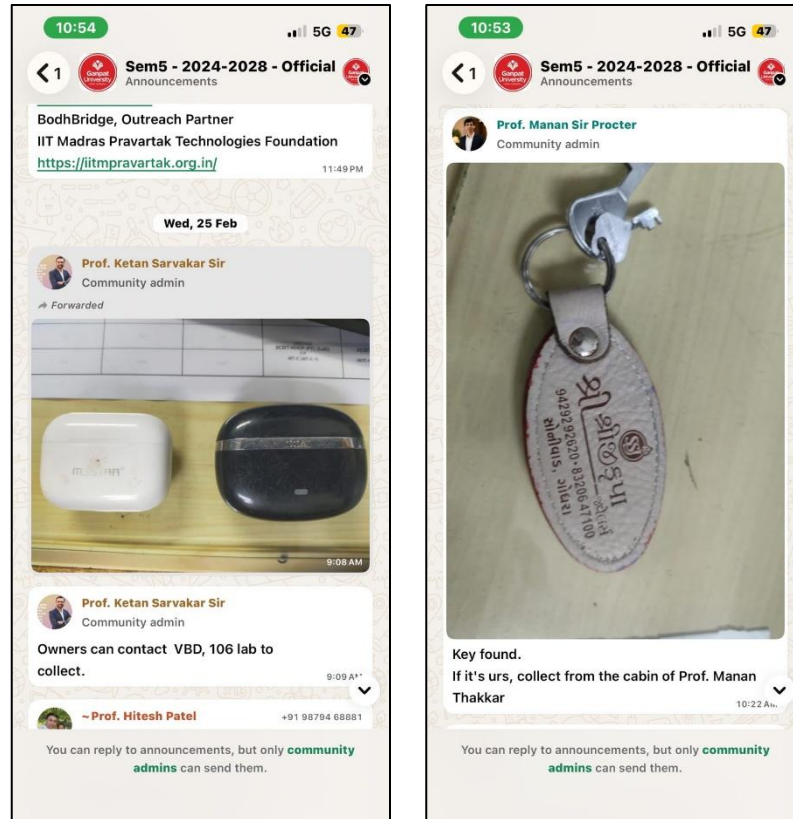
1.2 EXISTING SYSTEM

At present, there is no dedicated institutional platform for managing lost and found items on campus. When a student loses an item, the usual approach is to ask classmates, friends, faculty members, or other campus staff whether the item has been found. Information about a found item may also be communicated informally through personal messages or student communication groups. Similarly, when a person finds an item, it may be handed over to a nearby staff member or communicated informally to others. Since these activities are not managed through a common system, the information about a lost or found item may remain limited to the people who were directly involved in the communication.

The absence of a centralized platform makes it difficult for students to systematically search for items that may have been found elsewhere on campus. A student may need to contact different people or check different communication channels to determine whether an item has been recovered. At the same time, Staff or Admin personnel who receive a found item do not have a dedicated system through which its details, location, and current status can be recorded and made available for authorized users to view.

For example, if a student loses a wallet in the library, the student may initially ask nearby students or library personnel about it. If another person has already found the wallet and handed it over to a staff member, the student may not have a direct way to know that the item has been recorded or where it is currently being kept. The student may therefore need to make repeated enquiries until the information reaches them. FINDLY addresses this communication gap by providing a centralized record of lost and found items that can be searched through relevant item information. One can understand the situation from the below Snapshots showing how currently such situation is handled:





Above is the Existing System which is being followed to fulfill Lost & Found

The existing approach also does not provide a clearly defined role-based workflow. In an informal communication environment, information about a found item can potentially be shared by anyone, while there is no dedicated institutional record connecting the found item with the staff member responsible for its custody. FINDLY introduces a role-based approach in which **Students can report lost items and browse found-item records, while Staff and Admin users are responsible for logging found items and managing their custody and resolution.**

Thus, the main limitation of the existing system is not the absence of a physical register or notice board, but the **absence of a centralized, structured and role-controlled digital system** for managing campus lost-and-found information.

1.3 CHALLENGES OF EXISTING SYSTEM

The existing informal approach creates several challenges in managing lost and found items across campus:

- **No centralized record:** Information about lost and found items is distributed across personal communication, conversations and informal communication channels rather than being maintained in one dedicated system.
- **Difficulty in searching information:** Students do not have a structured search facility through which they can search available found items using details such as item name, category, location or date.
- **Dependence on informal communication:** Students may need to contact different people or check different communication channels to determine whether their lost item has been found.
- **No structured found-item logging:** There is no dedicated system through which authorized Staff or Admin users can record the details of an item that has been handed over to them.
- **Limited visibility of item status:** Without a centralized record, it can be difficult for a student to know whether an item is currently active, has been resolved, or is still being handled by the institution.
- **No centralized custody information:** Information about where a found item is currently being kept may not be available through a common platform.
- **Lack of role-based access:** The existing informal process does not provide a software-based distinction between the activities that should be performed by Students, Staff, and Admin users.
- **Limited administrative visibility:** There is no centralized dashboard through which authorized administrators can view overall lost-and-found activity and manage system records.

These challenges make the existing process dependent on communication between individuals rather than on a structured digital workflow. FINDLY is designed to address these limitations by providing a centralized platform with role-based access, searchable records, custody information, status tracking and administrative management.

1.4 PURPOSE

The purpose of FINDLY - Campus Lost & Found is to provide a centralized web-based platform for managing lost and found items within an academic campus. The system is designed to replace the existing informal communication-based approach with a structured process through which Students can report lost items and search available found-item records, while

Staff and Admin users can record found items, manage their custody information, and update their status.

The system also aims to establish clear role-based access for the three major user types: Student, Staff, and Admin. Students are provided with features related to reporting and searching for items, whereas Staff and Admin users are given additional capabilities for managing found items and their resolution. When a found item is handed over to its recipient, the authorized Staff or Admin user can record relevant recipient information such as name, contact number, and enrollment or employee number. FINDLY records this information as part of the item-resolution process; it does not technically guarantee or algorithmically determine ownership.

From the Student's perspective, the purpose of the system is to provide a single place where lost items can be reported and found-item records can be searched without depending entirely on informal communication. From the Staff perspective, the purpose is to provide a structured method for recording found items, maintaining custody information and updating their status. From the Admin perspective, the purpose is to provide centralized management of users, categories, item records and system-level information.

Overall, FINDLY aims to make the campus lost-and-found process more organized, searchable, transparent and easier to manage by bringing its major activities into a single role-based web platform.

1.5 SCOPE

The scope of FINDLY - Campus Lost & Found for Semester V, Capstone Project - I covers the analysis, design, database planning, UML modelling, high-fidelity Figma prototype and, subject to available development time, implementation of a web-based campus lost-and-found management system.

In Scope

- User registration and login for Student, Staff and Admin accounts.
- Role-based access control to provide different functions according to the user's role.
- Reporting of Lost Items by Students through a structured form.
- Logging of Found Items by Staff and Admin users only.
- Browsing of available Found Items by Students.
- Searching and filtering Found Items using relevant information such as keyword, category, date and location.
- Viewing detailed information about lost and found item records according to the user's permissions.

- Student management of their own Lost Item reports, including viewing, editing and deleting their reports where permitted.
- Staff/Admin management of Found Item records.
- Updating the custody location of Found Items by authorized Staff/Admin users.
- Marking an item as Claimed/Resolved when the item has been handed over.
- Recording recipient information during item handover, including name, contact number and enrollment number or employee number as applicable.
- Notification functionality for relevant system updates and status changes.
- Admin management of users and item categories.
- Admin management and oversight of Lost and Found records.
- Dashboard and summary information for authorized users.

Out of Scope

- Automatic ownership verification or a system-generated guarantee that a person is the genuine owner of an item.
- Student-created Found Item records.
- Student-submitted claim requests or online claim approval/rejection workflows.
- Online payment or reward processing.
- Physical delivery or courier services.
- Integration with the institution's official ERP or student database for automatic identity verification.
- AI-based image matching between Lost and Found Items.
- Native Android/iOS applications during the current project phase.

1.6 OBJECTIVES

- To design a centralized web-based platform for reporting, searching and managing lost and found items within the campus.
- To provide Students with a structured facility for reporting Lost Items and browsing Found Items.
- To restrict Found Item logging to authorized Staff and Admin users.
- To provide search and filtering facilities that allow users to locate relevant Found Item records using information such as keyword, category, date and location.
- To allow Students to view and manage their own Lost Item reports.
- To allow Staff and Admin users to maintain Found Item records, including custody location and item status.

- To provide a structured item-resolution process in which authorized Staff or Admin users can mark an item as Claimed/Resolved and record the recipient's relevant details.
- To provide role-based access so that Students, Staff and Admin users can access only the functions appropriate to their roles.
- To provide notifications for relevant item and account updates.
- To provide Admin users with centralized management facilities for users, categories and lost-and-found records.

1.7 ADVANTAGES

FINDLY provides the following advantages over the existing informal approach:

- **Centralized information:** Lost and found records are maintained in one dedicated platform instead of being distributed across informal communication channels.
- **Easy searching:** Users can search and filter available records using relevant item information.
- **Role-based access:** Different functionalities are provided to Students, Staff and Admin users according to their responsibilities.
- **Controlled Found Item logging:** Only Staff and Admin users can create Found Item records, reflecting their role in handling items handed over to the institution.
- **Custody information:** Authorized Staff/Admin users can record and update the location where a Found Item is being kept.
- **Status tracking:** The system can maintain the current status of an item, making it easier to understand whether an item is still active or has been resolved.
- **Structured handover record:** When an item is handed over, authorized Staff/Admin users can record the recipient's name, contact number and enrollment/employee number.
- **Improved administrative management:** Admin users can manage users, categories and item records through dedicated system functions.
- **Notifications:** Users can receive relevant updates through the notification module.
- **Better organization:** The system replaces scattered communication with a structured digital workflow for campus lost-and-found management.

1.8 EXISTING SYSTEM vs. PROPOSED SYSTEM

Aspect	Existing(Manual) System	Proposed System - FINDLY
Reporting a lost item	Usually communicated through friends, classmates, staff or informal communication channels	Structured Lost Item reporting form for Students
Recording a found item	Item may be communicated informally or handed over to a staff member without a centralized digital record	Found Item can be logged by authorized Staff/Admin users
Searching items	No centralized search facility; users depend on personal communication	Search and filter using keyword, category, date and location
Access Control	No dedicated software-based role separation	Role-based access for Student, Staff and Admin
Found-item custody	Custody information may depend on individual communication	Staff/Admin can record and update custody location
Item Status	Status may not be centrally visible	Item status can be maintained in the system
Student Reports	Difficult to organize and track multiple reports	Students can view and manage their own Lost Item reports
Item Handover	Handover information may not be maintained in a centralized digital record	Staff/Admin can record recipient name, contact number and enrollment /employee number
Notifications	No dedicated system notification mechanism	In-app notifications for relevant updates
Administration	No centralized digital management interface	Admin dashboard and management functions
User Management	No centralized role-based user management	Admin can manage users and roles
Category management	No standardized digital categorization	Admin can manage item categories

[TABLE 1.8.1 EXISTING vs. PROPOSED SYSTEM]

1.9 Generalized Diagram

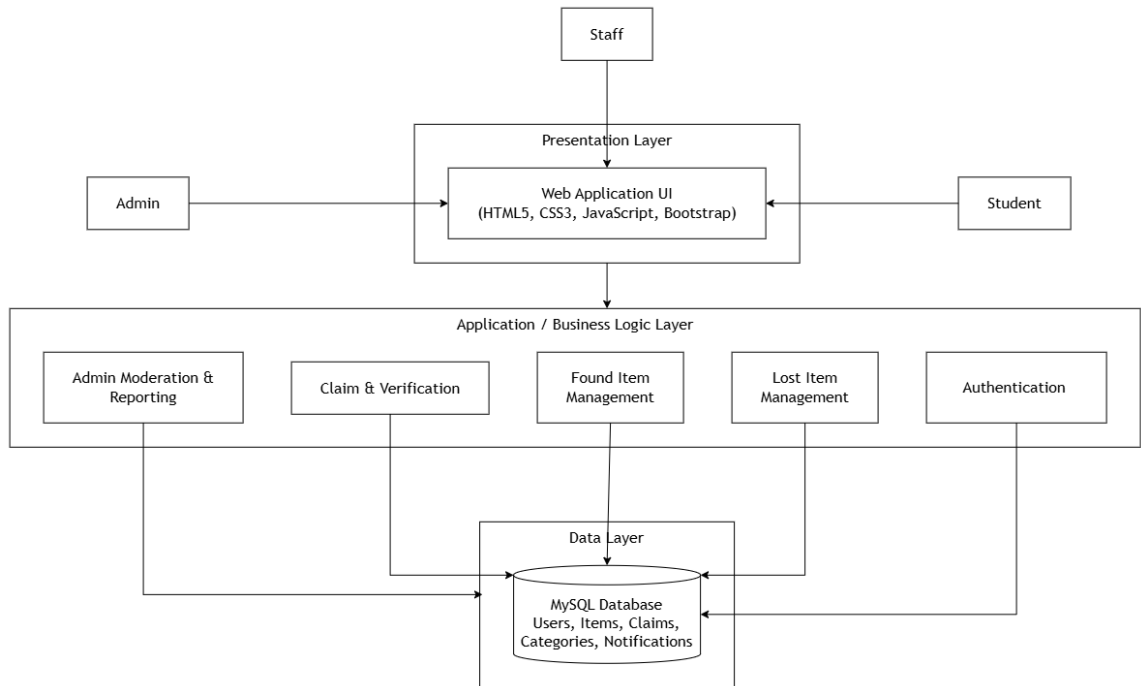


Figure 1.9.1 Generalized Diagram (Architecture)

The generalized architecture of FINDLY consists of three major layers: the Presentation Layer, Application Layer and Data Layer. The Presentation Layer provides the user interface through which Students, Staff and Admin users interact with the system. The interface is designed to provide role-specific navigation and functionality according to the logged-in user's permissions.

The Application Layer contains the main business logic of the FINDLY system. It handles functions such as user authentication, role-based access, lost-item reporting, found-item logging, searching and filtering, item management, custody updates, notifications and item resolution. The application layer also ensures that Found Item records can only be created and managed by authorized Staff and Admin users.

The Data Layer is responsible for storing the system's structured information, including user details, item records, categories, notifications and relevant status information. A relational database such as MySQL can be used to maintain these records. The separation of the presentation, application and data layers provides a structured architecture in which the user interface, application functionality and database can be developed and maintained as separate logical components.

2. LITERATURE SURVEY

2.1 INTRODUCTION

Before finalising the design of FINDLY - Campus Lost & Found, ten existing research papers, project reports and lost-and-found platforms were reviewed. The purpose of the literature survey was to understand how existing systems approach the reporting, searching, matching and management of lost and found items, the technologies adopted by these systems, and the limitations that remain in practical campus environments. The review also helped identify design considerations relevant to FINDLY, particularly the need for a centralised searchable platform and appropriate control over who can record found items within an institutional environment.

The reviewed systems demonstrate that digital lost-and-found platforms can improve upon informal communication by providing structured records, search facilities and better visibility of reported items. However, many systems are designed around a general user model in which users can report both lost and found items. FINDLY follows a different operational approach suited to the project's campus environment: Students can report items that they have lost and can browse or search items that have been found, whereas Found-item records can be created only by authorised Staff or Admin users. This reflects the fact that a found item is expected to be physically handed over to an authorised person or institutional location before it is entered into the system.

Another important observation from the reviewed systems is that the final physical handover of an item is often outside the scope of the software. FINDLY therefore focuses on maintaining a structured record of the handover rather than attempting to provide automatic ownership verification. When a found item is collected, authorised Staff/Admin personnel can record the recipient's name, contact number and enrollment or employee number and update the item's status accordingly. This provides an institutional record of the handover while keeping the actual physical identification and release of the item under the responsibility of authorised personnel.

2.2 REVIEWED PAPERS / PROJECT

2.2.1 *QR Code Item Registration (2024)*

Sinha et al. proposed an approach for improving campus lost-and-found services by converting personalised item-registration information into QR codes. When a finder scans the QR code associated with an item, the system can direct the finder towards the corresponding owner's listing. The primary

objective of the approach is to reduce the time required to establish a connection between a lost item and its owner once the item has been identified.

The approach demonstrates how QR-based identification can simplify the recovery process, particularly when an item already carries an associated QR code. However, it is primarily oriented towards direct interaction between the finder and the owner. It does not focus on an institutional custody workflow in which found items are first recorded by authorised campus personnel. FINDLY therefore adopts a different approach by making Staff/Admin responsible for creating Found-item records.

2.2.2 UniFind (2025)

The UniFind project proposed an intelligent lost-and-found platform using artificial intelligence techniques such as image recognition and natural language processing to identify similarities between Lost and Found item descriptions. The system aims to reduce the effort required for users to manually search through large numbers of listings by automatically identifying potentially related items.

The major contribution of UniFind is its automated matching capability. Such an approach can become useful when a large number of reports are available, although the reliability of automated image and text matching remains an important consideration. FINDLY does not depend on AI-based matching in its current scope. Instead, it provides structured keyword, category, date and location-based search facilities, keeping the current system simpler and more appropriate for the project's initial implementation.

2.2.3 UKM Campus Lost-and-Found System (2023)

A case study conducted at Universiti Kebangsaan Malaysia developed and evaluated a web/mobile lost-and-found system for students and staff. The system focused on providing a centralised mechanism for reporting and searching lost-and-found information rather than depending entirely on informal communication methods. Usability and user acceptance were also considered through pretesting involving campus users.

The study demonstrates the usefulness of a centralised digital interface for campus lost-and-found management. However, the study was conducted within a specific institutional environment and primarily focused on reporting, searching and usability. The reviewed material does not establish the same Staff/Admin-controlled Found-item logging approach used by FINDLY.

FINDLY incorporates role-based access according to the responsibilities of different campus users.

2.2.4 Peer-to-Peer Matching System

A peer-to-peer approach to automated lost-and-found management reviewed several RFID-based and application-based methods for connecting people who lose items with people who find them. These approaches generally attempt to make reporting faster and facilitate direct communication between users.

Such systems demonstrate the benefits of immediate digital reporting and automated matching. However, a peer-to-peer model may not correspond directly to an institutional environment where found belongings are expected to be handed over to a responsible staff member or designated campus authority. FINDLY therefore separates the responsibilities of reporting a Lost item and recording a Found item, with Found-item creation restricted to Staff/Admin accounts.

2.2.5 College Lost and Found Management System (2026)

A college-focused web-based Lost and Found Management System examined the limitations of manual and fragmented approaches to campus lost-and-found management. The proposed system centralised reporting, searching and recovery information through a dedicated web portal.

The project highlights the importance of maintaining a structured digital record as the number of students, locations and reported items increases. Its approach supports the centralised model adopted by FINDLY. However, the reviewed system does not provide the same emphasis on restricting Found-item creation to authorised campus personnel and maintaining a structured record of the eventual recipient during physical handover.

2.2.6 FoundLost (2024)

FoundLost is a college-oriented Lost and Found website developed using HTML, CSS, JavaScript and jQuery on the front end, with Python/Django and MySQL forming the back-end technology stack. The system aims to automate the reporting and retrieval process and replace fragmented communication with an online platform.

The project demonstrates the practical feasibility of using conventional web technologies for a campus Lost and Found application. The reviewed work also

identifies usability considerations for first-time users and suggests future improvements such as notifications and multilingual support. FINDLY similarly uses a structured web interface and includes an in-app notification mechanism so that users can receive updates regarding relevant item records and status changes.

2.2.7 Eye-Tracking Usability Study (2026)

A 2026 usability study evaluated an online Lost and Found system for a higher-education environment using an eye-tracking-based evaluation approach. The study focused on understanding how users interact with the system and assessing the usability and satisfaction associated with its functionality.

The study is particularly relevant to FINDLY because a campus Lost and Found platform must not only provide the required functions but must also remain simple for students and institutional personnel to use. However, the primary focus of the study was usability evaluation rather than the detailed management of institutional custody and physical handover. FINDLY therefore incorporates a straightforward role-based interface in which users are presented only with functions relevant to their responsibilities.

2.2.8 Comparative Reliability Study

A comparative IEEE study evaluated the reliability of Lost and Found management systems across multiple educational institutions, including Universiti Tunku Abdul Rahman, California State Polytechnic University and Nazareth School of National University. The study examined the reliability and operational characteristics of different approaches to digital Lost and Found management.

The comparison indicates that the reliability of a Lost and Found platform depends not only on whether information is stored digitally but also on how the system controls and manages user interactions. This observation is relevant to FINDLY's role-based design. Rather than allowing every account to perform the same actions, FINDLY separates Student and Staff/Admin responsibilities, particularly for the creation and management of Found-item records.

2.2.9 VNR VJIET Lost and Found System (2022)

Vasavi described a Lost and Found system developed for VNR VJIET using a mobile application, Firebase, Spring Boot and PostgreSQL. The system was

intended to provide students with a digital mechanism for reporting and managing information about lost and found belongings.

The project demonstrates the use of mobile and cloud-oriented technologies for campus Lost and Found management. It also identifies practical difficulties associated with item information and location accuracy. FINDLY takes a simpler web-based approach for the current project phase and emphasises structured item information, including category, location, date and description, so that users can search available records more effectively.

2.2.10 Find Mine: Find the Lost Items via Mobile App (2021)

Choudhary et al. presented "Find Mine: Find the Lost Items via Mobile App", a mobile application designed to help users report and search for lost belongings. The system provides search facilities using information such as keywords and categories, offering an improvement over approaches based entirely on notice boards or informal communication.

The project demonstrates the usefulness of searchable digital records in reducing the effort required to locate information about lost items. However, the final physical handover process remains largely outside the application's core workflow. FINDLY similarly recognises that the physical collection of an item occurs outside the web application, but provides Staff/Admin functionality for recording the recipient details and updating the item's status after handover.

2.3 COMPARATIVE ANALYSIS

Paper / Report	Purpose	Major Features	Technology Used	Limitations
[1] QR Code Item Registration (2024)	Speed up identification of lost items	QR-coded item registration; image-embedded QR codes	Web application; QR generation	Primarily focused on direct finder-owner interaction; no institutional custody workflow
[2] UniFind (2025)	Automate Lost/Found matching	AI image recognition; NLP matching; cloud storage	AI/ML; cloud services	Matching reliability and data-security concerns
[3] UKM Campus System (2023)	Centralise campus lost-and-found reporting	Web/mobile reporting; keyword and category search;	Web + mobile	Evaluated within a single campus environment; limited focus on

Paper / Report	Purpose	Major Features	Technology Used	Limitations
		usability evaluation		institutional verification
[4] P2P Matching System	Facilitate direct finder-owner matching	RFID/application-based reporting; peer matching	RFID; mobile application	Peer-to-peer approach does not emphasise institutional custody
[5] College Lost & Found Management System (2026)	Centralise campus lost-and-found management	Online reporting; search; recovery	Web-based portal	Limited emphasis on Staff/Admin-controlled found-item workflow
[6] FoundLost (2024)	Automate reporting and retrieval	Web reporting; retrieval workflow	HTML, CSS, JavaScript, jQuery, Django, MySQL	Usability considerations; notifications identified as an area for improvement
[7] Eye-Tracking Usability Study (2026)	Evaluate usability of an online lost-and-found system	Reporting; searching; usability evaluation	Web application + eye-tracking study	Primarily focused on user experience rather than institutional custody
[8] Comparative Reliability Study	Compare reliability across institutions	Cross-institution reliability evaluation	Comparative study	Highlights importance of controlled system operation and user permissions
[9] VNR VJIET System (2022)	Provide mobile lost-and-found reporting and search	Mobile reporting; item information; location support	Firebase; Spring Boot; PostgreSQL	Location and operational challenges; limited institutional workflow
[10] Find Mine (2021)	Improve lost-item searching	Keyword and category search; mobile interface	Mobile application	Final physical handover remains outside the digital workflow

[Table 2.3 COMPARATIVE ANALYSIS]

2.4 SUMMARY OF FINDINGS

The literature survey indicates that digital Lost and Found systems can significantly improve the accessibility and organisation of information compared with fragmented communication methods. Across the reviewed

systems, centralised reporting, searchable records and structured item information are recurring features that help users locate relevant Lost and Found information more efficiently. These findings support the decision to provide FINDLY with a single searchable platform rather than relying on independent messages, conversations or informal communication between users.

A second important observation is that existing systems use different approaches to determine who can create and manage Found-item records. Peer-to-peer systems generally allow users to report found belongings directly, whereas an institutional environment may require greater control over the creation of Found-item records. FINDLY therefore adopts a role-based approach in which Students can report Lost items, while only Staff/Admin accounts can log Found items. This reflects the operational responsibility of authorised campus personnel who receive or hold found belongings.

The reviewed systems also show that the final physical recovery of an item is commonly an activity that occurs outside the software. FINDLY does not attempt to provide automatic ownership verification or guarantee that a person is the genuine owner through software alone. Instead, once a person collects a Found item, authorised Staff/Admin personnel can record the recipient's name, contact number and enrollment or employee number and update the item's status. This creates a structured record of the handover while keeping physical identification and release under the responsibility of authorised personnel.

Based on the overall findings, FINDLY focuses on a practical and manageable set of features for the current project phase: centralised Lost and Found records, role-based access, Student Lost-item reporting, Staff/Admin Found-item logging, searchable Found-item listings, management of reported Lost items, notifications and structured recording of item handovers. More advanced capabilities such as AI-based image matching, automatic integration with institutional databases and native mobile applications are considered outside the current scope and may be explored as future enhancements.

3. TECHNOLOGICAL SPECIFICATIONS

FINDLY - Campus Lost & Found is designed as a web-based application following a three-tier architecture consisting of the presentation layer, application layer and data layer. The presentation layer provides the user interface through a web browser, the application layer manages authentication, role-based access, item reporting, searching, item status and handover-related operations, while the data layer provides persistent storage for users, lost and found items, categories, notifications and related records. The selected technologies are intended to provide a practical, low-cost and maintainable solution suitable for a campus-level system.

3.1 FRONT-END TECHNOLOGIES

Html5:

- HTML5 is used to provide the basic structure of the FINDLY web application. It defines the structure of pages such as the landing page, login and registration pages, Student dashboard, Lost Item reporting form, My Reports page, Found Items listing, item-detail pages, Staff/Admin pages and profile-related screens.
- Semantic HTML elements are used wherever appropriate to organise page content and improve maintainability. HTML forms are used for collecting information such as lost-item details, user registration information and recipient details recorded during the handover of a found item.

Css3:

- CSS3 is used to control the visual appearance and layout of the FINDLY interface. It is responsible for page spacing, typography, colours, buttons, cards, navigation panels, tables, forms and responsive layouts.
- A consistent visual theme is maintained across the different modules so that Student, Staff and Admin interfaces remain recognisable as parts of the same system. CSS is also used to provide responsive behaviour so that the interface can be accessed from different screen sizes.

Bootstrap 5:

- Bootstrap 5 can be used to support responsive page layouts and commonly required interface components. Its grid system and utility classes simplify the development of dashboards, forms, navigation elements, tables and responsive layouts.

- Bootstrap also reduces the amount of custom CSS required for common interface structures and helps maintain consistency across the different pages of FINDLY.

JavaScript / jQuery:

- JavaScript is used to provide client-side interactivity within FINDLY. It can be used for form validation, search and filter interactions, confirmation dialogs, dynamic interface updates and other client-side operations.
- jQuery may be used where simplified DOM manipulation and AJAX-based communication are required. For example, JavaScript/jQuery can support dynamic filtering of Found Items and communication with server-side PHP handlers without requiring unnecessary page reloads.
- The front end may also use role information to display appropriate navigation options. For example, the **Log Found Item** functionality is available to Staff and Admin accounts and is not provided to Student accounts. This front-end restriction is complemented by server-side access control so that permissions are not dependent only on the visibility of a button.

3.2 BACK-END TECHNOLOGIES**PHP:**

PHP is used as the server-side programming language for FINDLY. It manages the core application logic and communicates between the front-end interface and the MySQL database.

The PHP layer handles functions such as:

- User registration and login
- Session management
- Role-based access control
- Lost-item reporting by Students
- Found-item logging by Staff/Admin
- Editing and deletion of Student's own reports
- Searching and filtering of items
- Updating item status
- Recording recipient information during item handover
- Notification generation
- Dashboard statistics
- Category and user-related administrative operations

PHP is suitable for the project because it is widely used for web application development, integrates directly with MySQL and can be deployed using commonly available Apache-based hosting environments.

MYSQL:

MySQL is used as the relational database management system for FINDLY. It provides persistent storage for the application's structured data.

The database is expected to contain information related to:

- Users
- Categories
- Lost Items
- Found Items
- Notifications
- Recipient/Handover records
- Other supporting records required by the final implementation

Relationships between database tables are maintained using primary keys and foreign keys wherever applicable. Appropriate constraints are used to maintain data consistency.

Indexes can be applied to frequently searched or filtered attributes such as item status, category, date and location to improve retrieval performance as the number of records increases.

Apache / XAMPP:

Apache is used as the web server during development, with XAMPP providing the local development environment for the PHP and MySQL components of FINDLY.

XAMPP allows the project team to run the PHP application locally and connect it with the MySQL database during development and testing. phpMyAdmin, included within the XAMPP environment, can be used to create, modify and inspect the database structure and stored records.

3.3 SUPPORTING TOOLS AND LIBRARIES

The following tools support the development, design, testing and collaboration activities of the FINDLY project.

Tool / Library	Purpose
Bootstrap Icons / Font Awesome	Icons used across navigation, buttons, notifications, profiles and item-related actions
Git & GitHub	Version control and team collaboration
Figma	High-fidelity UI/UX prototyping and interactive page navigation (Chapter 7)

Tool / Library	Purpose
phpMyAdmin	Database administration, schema management and data inspection during development
Postman	Manual testing of PHP endpoints/AJAX handlers during development

[Table:3.3.1 SUPPORTING TOOLS AND LIBRARIES]

3.4 WHY THIS TECHNOLOGY STACK?

- The selected technology stack is appropriate for a campus-level web-based lost-and-found system because it provides the required functionality without introducing unnecessary complexity.
- **PHP, MySQL and Apache/XAMPP** provide a practical foundation for implementing the server-side application and relational database. These technologies are widely supported and can be operated in a local development environment with relatively low infrastructure requirements.
- The use of **HTML5, CSS3, Bootstrap and JavaScript/jQuery** allow the project team to develop a consistent and responsive interface for Students, Staff and Admin users. The technologies are also suitable for implementing forms, dashboards, searchable listings, filters and role-specific navigation within the available project timeframe.
- **MySQL** provides a structured approach for maintaining the centralised record of lost and found items. This is particularly important for FINDLY because the system depends on being able to store, retrieve, search and update item information instead of relying on scattered informal communication.
- **Git and GitHub** support collaborative development by allowing team members to maintain versions of the project and coordinate changes. **Figma** supports the design and demonstration of the working prototype before or alongside implementation.

4. FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS

This chapter defines the four core functional modules of FINDLY and describes the functional requirements of each module using the Input-Process-Output format. It also establishes the role-based permissions applicable to Student, Staff and Admin accounts, followed by the system's non-functional requirements. The requirements are structured according to the responsibilities of each user role and the workflow through which Lost and Found items are reported, managed, searched and resolved.

4.1 Module 1 - Authentication & Account Module (Guest / Student / Staff / Admin)

The Authentication & Account Module provides the basic account-related functionality required to access FINDLY. It manages user registration, authentication and logout while ensuring that access to system features is determined according to the user's assigned role.

4.1.1 Register

Input:

- Full name
- Institute email
- Contact number
- Password
- Confirm password
- Relevant identification information, such as enrollment number where applicable

Process:

- The system validates the entered information and checks whether the email address is already registered.
- Password and confirmation fields are checked for a valid match and required format.
- The account information is stored in the database with the appropriate account status and role according to the system's registration policy.

Output:

- A successful registration message is displayed.
- The user is redirected to the Login page.
- Appropriate validation messages are displayed if the submitted information is invalid or already registered.

4.1.2 Login

Input:

- Registered email

- Password

Process:

- The system searches for the account associated with the submitted email.
- The entered password is verified against the stored password.
- The system checks whether the account is active and determines the user's role.

Output:

- On successful authentication, the user is redirected to the appropriate dashboard based on their role.
- If authentication fails, an appropriate error message is displayed and access is denied.

4.1.3 Logout**Input:**

- User selects the Logout option.

Process:

- The system terminates the user's active session.
- Session-related authentication information is cleared.

Output:

- The user is redirected to the Login page.

4.2 Module 2 - Student Module

The Student Module provides the functionality required by students to report items that they have lost and to search the Found Items recorded by Staff/Admin. Students can manage their own Lost-item reports but cannot create Found-item records or perform Staff/Admin handover operations.

4.2.1 Report Lost Item**Input:**

- Item name/title
- Category
- Description
- Last-seen location
- Date lost
- Optional item image

Process:

- The system validates the required information.
- The submitted Lost Item is associated with the currently logged-in Student account.
- The record is stored in the database with its appropriate status.

Output:

- A confirmation message is displayed.
- The submitted Lost Item appears in the student's My Reports section.
- The current status of the report is displayed.

4.2.2 *Browse Found Items*

Input:

- User selects the Found Items/browse option.

Process:

- The system retrieves available Found Item records.
- Records are displayed according to their current status and visibility.

Output:

- A list/grid of available Found Items is displayed.
- Each listing provides basic information and a View option for accessing its details.

4.2.3 *Search and Filter Items*

Input:

- Keyword
- Category
- Date
- Location
- Item type, where applicable

Process:

- The system applies the selected search criteria to the available item records.
- Matching records are retrieved from the database.

Output:

- Matching items are displayed.
- If no item satisfies the selected criteria, a suitable No Results Found message is displayed.

4.2.4 *View Item Details*

Input:

- User selects the View button associated with an item.

Process:

- The system retrieves the complete information permitted for the selected item.
- The system displays the item's details without exposing restricted Staff/Admin or recipient information.

Output:

- Detailed item information is displayed, including relevant information such as title, category, description, location, date and image where available.

4.2.5 View My Reports

Input:

- Student selects My Reports.

Process:

- The system retrieves Lost Item reports associated with the logged-in Student account.
- Each report is displayed along with its current status.

Output:

- The Student can view their submitted Lost Item reports and their current status.

4.2.6 Edit Own Report

Input:

- Selection of one of the Student's own Lost Item reports.
- Updated report information.

Process:

- The system verifies that the logged-in Student is the owner of the selected report.
- Permitted information is updated in the database.

Output:

- The updated report information is displayed.
- A confirmation message indicates that the report has been updated successfully.

4.2.7 Delete Own Report

Input:

- Selection of the Student's own Lost Item report.
- Confirmation of deletion.

Process:

- The system verifies that the logged-in Student is associated with the selected report.
- The report is removed or deactivated according to the system's deletion mechanism.

Output:

- The report is no longer displayed among the Student's active reports.
- A confirmation message is displayed.

4.2.8 View Notifications

Input:

- Student selects the notification option.

Process:

- The system retrieves notifications associated with the logged-in Student.
- Notifications are displayed according to their date and read/unread status.

Output:

- The Student can view relevant notifications regarding their reports and system status changes.

4.2.9 View Profile

Input:

- Student selects the Profile option.

Process:

- The system retrieves the account information associated with the logged-in Student.

Output:

- The Student's profile information is displayed.

4.3 MODULE 3 - STAFF MODULE

The Staff Module is designed for users responsible for handling Found Items within the institution. Staff can record Found Items, maintain custody information and complete the item handover process. Staff do not have access to administrative functions such as user and category management.

4.3.1 Log Found Item

Input:

- Item name/title
- Category
- Description
- Location where the item was found
- Date found
- Current custody location
- Optional item image

Process:

- The system verifies that the logged-in account has Staff or Admin privileges.
- Required information is validated.

- A Found Item record is created and associated with the Staff/Admin account that logged it.

Output:

- The Found Item is successfully recorded.
- A confirmation message is displayed.
- The item becomes available according to its current system status.

4.3.2 Update Found Item / Custody Information**Input:**

- Selection of a Found Item
- Updated custody location or relevant status information

Process:

- The system verifies that the user has permission to modify the selected Found Item.
- The updated custody information is stored in the database.

Output:

- The updated custody information is displayed on the item's details.
- The current status/location of the Found Item is updated.

4.3.3 Mark Item as Claimed**Input:**

- Selection of an available Found Item
- Confirmation that a person has approached Staff/Admin to collect the item

Process:

- Staff/Admin performs the required verification in person.
- The Staff/Admin checks the available information related to the item and the person collecting it.
- If the handover process is accepted, the item's status is changed to Claimed.

Output:

- The Found Item status changes to Claimed.
- The item proceeds to the recipient-detail recording stage.

4.3.4 Record Recipient Details**Input:**

- Recipient full name
- Recipient contact number
- Recipient enrollment number / employee number

Process:

- The system verifies that the logged-in account is Staff or Admin.

- The recipient information is recorded against the relevant Found Item.
- The information is stored as part of the item's handover record.

Output:

- Recipient details are successfully recorded.
- The information is available to authorised Staff/Admin users for future reference.

4.3.5 *Mark Item as Resolved*

Input:

- Selection of a Claimed Found Item
- Confirmation that the handover process has been completed

Process:

- The system checks that the required recipient information has been recorded.
- Staff/Admin confirms completion of the handover.
- The item's status is changed to Resolved and the relevant timestamp is recorded.

Output:

- The item status changes to Resolved.
- The item is removed from the active Found Item workflow and remains available in relevant history/statistics.

4.3.6 *View Limited Dashboard Statistics*

Input:

- Staff selects the Dashboard option.

Process:

- The system retrieves statistics permitted for the Staff role.
- The information is limited according to the role-permission matrix.

Output:

- Staff can view relevant activity/statistics.
- Administrative controls and institution-wide management functions remain unavailable.

4.3.7 *View Notifications*

Input:

- Staff selects the notification option.

Process:

- The system retrieves notifications associated with the Staff account.

Output:

- Relevant notifications and status updates are displayed.

4.3.8 *View Profile*

Input:

- Staff selects the Profile option.

Process:

- The system retrieves the Staff account information.

Output:

- Staff profile information is displayed.

4.4 Module 4 - Admin Module

The Admin Module provides administrative control over the FINDLY platform. Admin users have access to the operational functions available to Staff as well as additional functions for managing users, categories and overall system information.

4.4.1 *Moderate Pending Posts*

Input:

- Selection of a pending Lost or Found Item
- Approve or Reject decision

Process:

- The Admin reviews the submitted item information.
- The system updates the item's status according to the Admin's decision.
- The relevant decision and timestamp may be recorded for system tracking.

Output:

- An approved item becomes available according to the system's listing workflow.
- A rejected item remains unavailable for normal listings.
- The relevant user can be notified of the decision.

4.4.2 *Manage Categories*

Input:

- Category name
- Create, edit or delete action

Process:

- The system verifies Admin privileges.
- For a new category, the system checks whether the category already exists.
- The selected category operation is performed according to database constraints.

Output:

- The updated category list is displayed.

- Updated categories become available in relevant item forms and search filters.

4.4.3 *Manage Users*

Input:

- Selected user account
- Account-management action

Process:

- The system verifies Admin privileges.
- The selected account is updated according to the permitted administrative action.

Output:

- The user's account status/access is updated.
- The change takes effect according to the system's access-control rules.

4.4.4 *View Dashboard and Statistics*

Input:

- Admin selects the Dashboard option.

Process:

- The system retrieves and aggregates relevant system-level information.
- Statistics are generated from the stored Lost and Found records and user information.

Output:

- The Admin Dashboard displays summary statistics and relevant recent activity.

4.4.5 *Manage Found-Item Handover*

Input:

- Selection of a Found Item undergoing the handover process.

Process:

- Admin can perform the same Staff-level operational actions available for Found Items.
- The Admin can update the item's custody information, record recipient details and complete the item's status transition.

Output:

- The Found Item record reflects the latest custody and handover information.
- The item can be marked as Resolved after the required information has been recorded.

4.4.6 View Notifications

Input:

- Admin selects the notification option.

Process:

- The system retrieves notifications relevant to the Admin account.

Output:

- Relevant notifications and system updates are displayed.

4.4.7 View Profile

Input:

- Admin selects Profile.

Process:

- The system retrieves the Admin account information.

Output:

- Admin profile information is displayed.

4.5 Non-Functional Requirements

Non-functional requirements define the quality, security, usability, performance and maintainability characteristics expected from FINDLY.

NFR-01 - Security

- All user passwords shall be stored using an appropriate salted password-hashing mechanism. Plain-text passwords shall not be stored in the database or application logs.

NFR-02 - Input Validation

- All user-provided information shall be validated before being processed or stored. Database queries shall use appropriate parameterisation to reduce the risk of SQL injection and other input-based attacks.

NFR-03 - Role-Based Access Security

- The system shall enforce role-based permissions on the server side. Functions restricted to Staff/Admin, such as Found-item logging and handover management, shall not be accessible through direct requests from Student accounts even when the corresponding interface option is unavailable to them.

NFR-04 - Performance

- Under normal expected system load, common operations such as searching and filtering item records should provide results within an acceptable response time.

NFR-05 - Usability

- The interface shall be simple and understandable for Students, Staff and Admin users. Common operations such as reporting a Lost Item, searching Found Items and logging a Found Item should require a minimal number of steps.

NFR-06 - Responsive Design

- The web interface shall remain usable across desktop, tablet and mobile screen sizes. The responsive design shall accommodate smaller mobile viewports without making the major system functions inaccessible.

NFR-07 - Reliability

- The system shall maintain consistent item and user records during normal operation. Changes to item status and handover information shall be stored correctly so that the latest state of an item can be retrieved when required.

NFR-08 - Session Security

- Authenticated user sessions shall be managed securely and shall expire according to the application's defined session policy.

NFR-09 - File Upload Security

- Where item images are uploaded, the system shall restrict uploads to supported image formats and enforce an appropriate file-size limit. Unsupported or invalid files shall be rejected.

NFR-10 - Maintainability

- The application shall follow a consistent project structure and coding practice so that individual modules can be maintained and extended without unnecessarily affecting unrelated functionality.

NFR-11 - Scalability

- The database structure shall be designed to accommodate increasing numbers of users and Lost/Found Item records without requiring a complete redesign of the database structure.

NFR-12 - Privacy

- Recipient information collected during the physical handover process, including name, contact number and enrollment/employee number, shall be accessible only to authorised Staff/Admin users and shall not be displayed in public item listings.

4.6 Summary of Functional Requirements

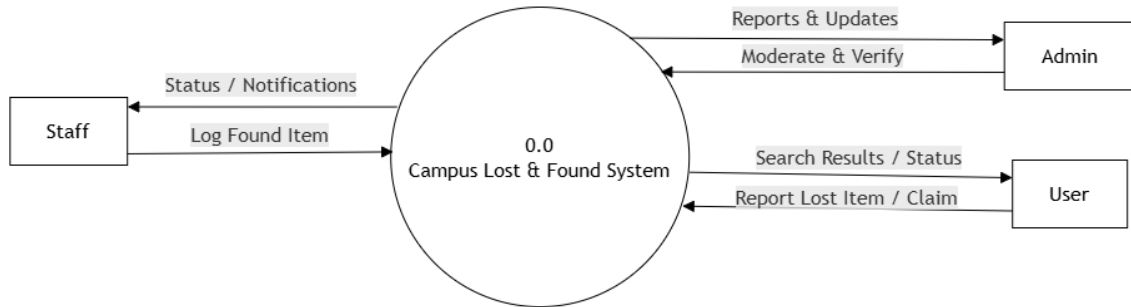
FINDLY is organised into four functional modules that collectively support the complete campus Lost and Found workflow. The Authentication & Account Module provides controlled access to the system, while the Student Module allows users to report Lost Items and search available Found Items. The Staff Module manages the recording and custody of Found Items and supports the physical handover process through recipient-detail recording. The Admin Module provides administrative oversight, including user and category management, dashboard statistics and the ability to perform Staff-level Found-item operations.

The role-based structure ensures that the responsibilities of Students, Staff and Admin users remain clearly separated. Students interact primarily with Lost-item reporting and searching, while Staff and Admin users manage Found Items and physical handovers. The system therefore provides a structured digital record of the Lost and Found process without claiming to replace the physical judgement involved in verifying a person at the time of item collection.

5. DIAGRAMS

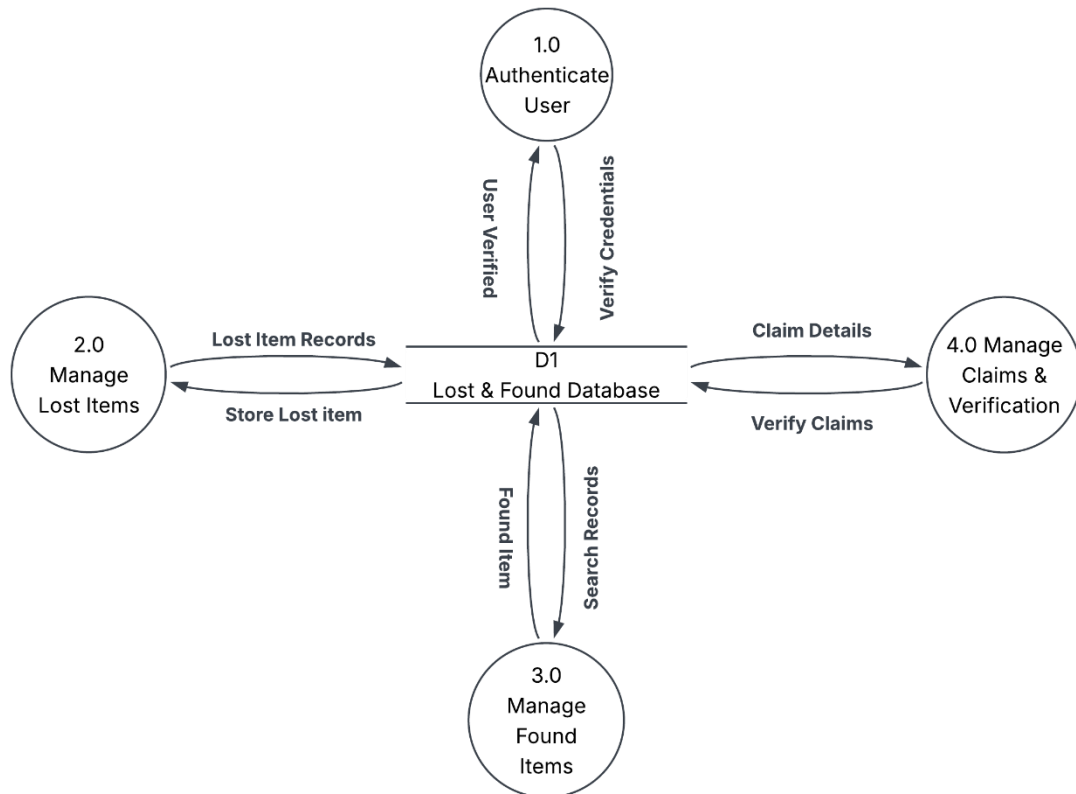
5.1 FLOW DIAGRAM (DATA FLOW DIAGRAMS)

5.1.1 Context Level (DFD Level-0)



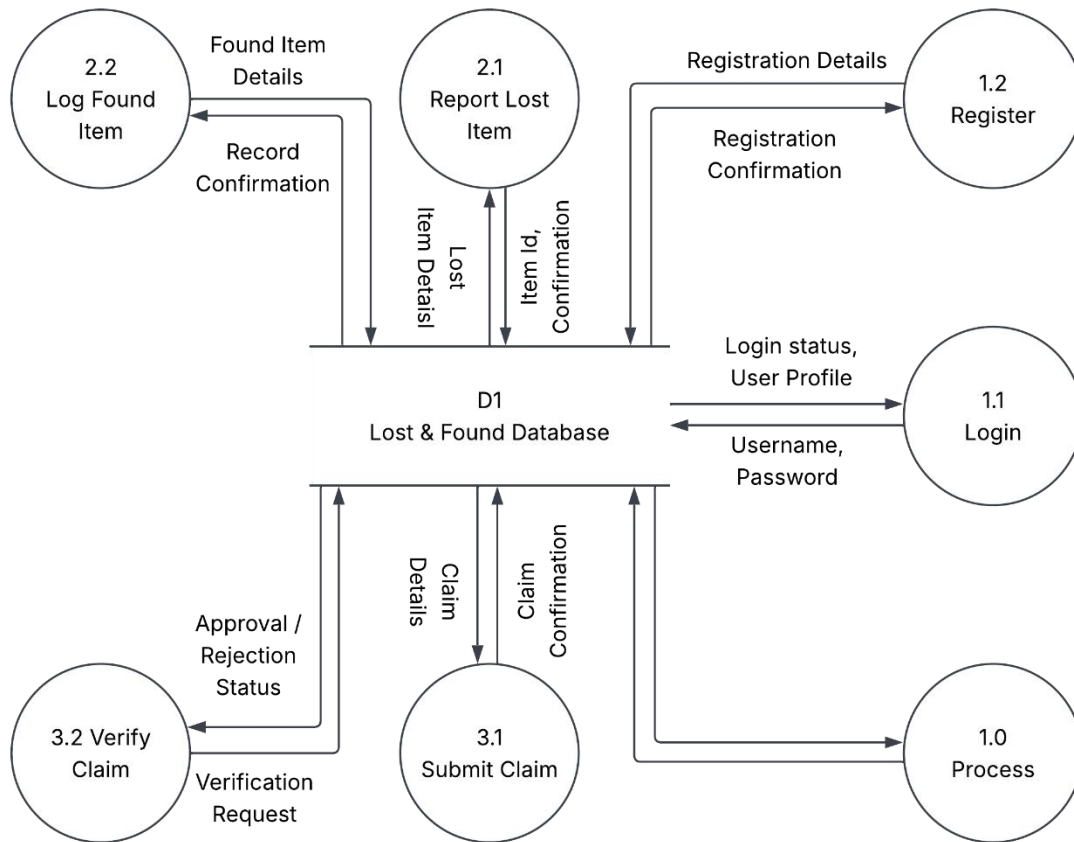
[Figure 5.1.1 DFD Level-0]

5.1.2 Data Flow Diagram Level-1



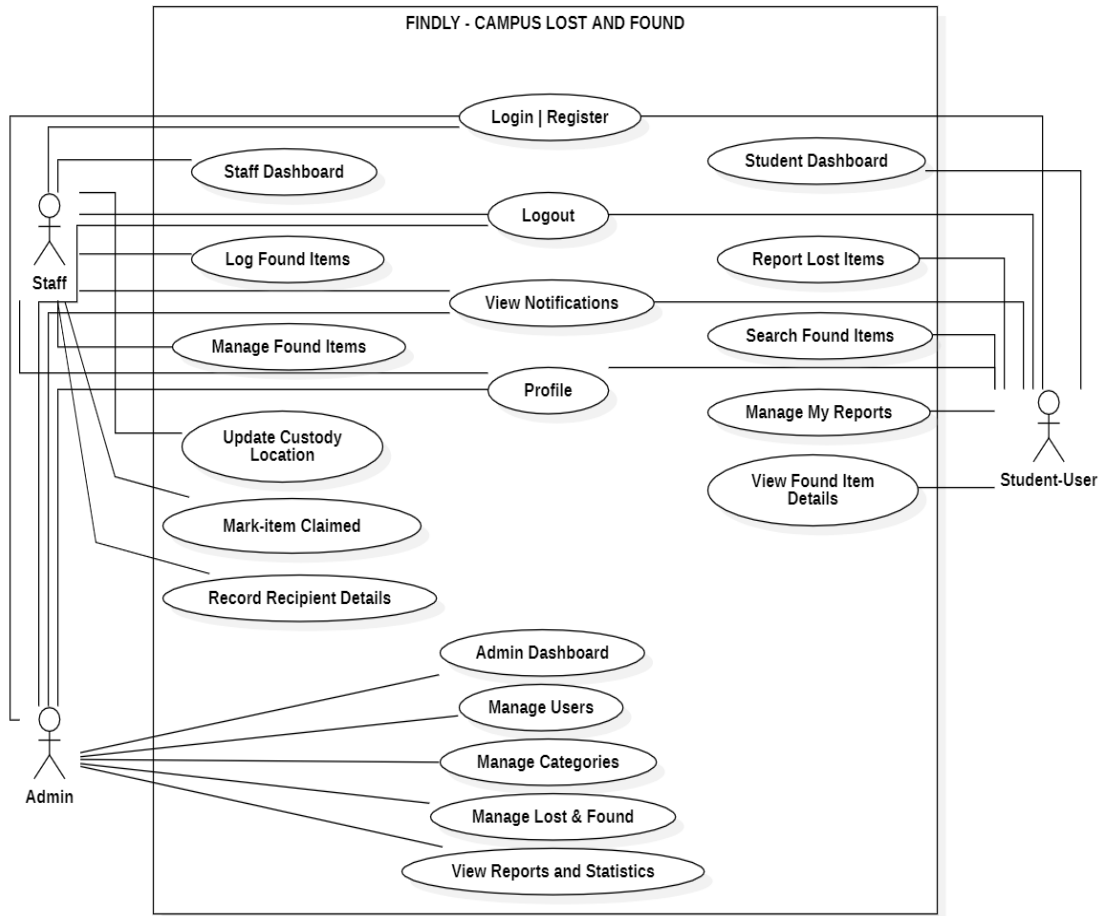
[Figure 5.1.2 DFD Level-1]

5.1.3 Data Flow Diagram Level-2



[Figure 5.1.3 DFD Level-2]

5.2 Use Case Diagram



[Figure 5.2.1 Use Case Diagram]

5.3 Activity Diagram

5.3.1 Student / User

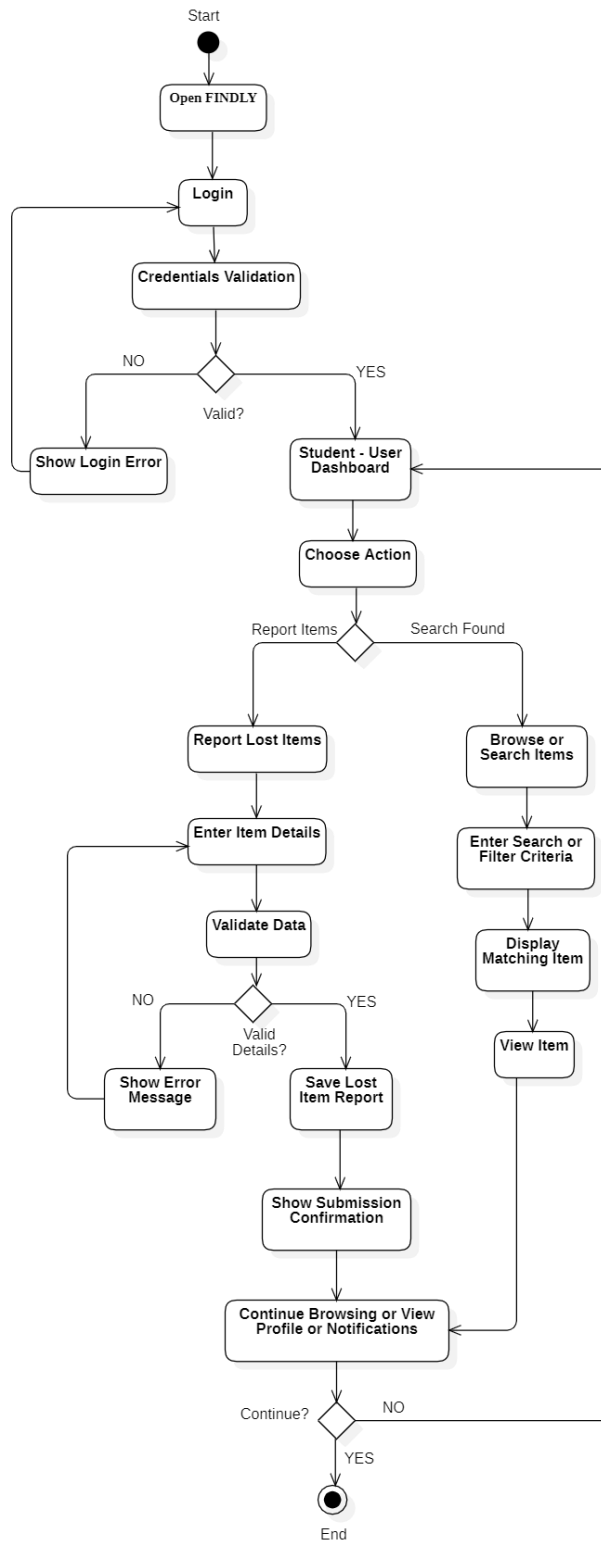


Figure 5.3.1.1 Activity- Student / User

5.3.2 Staff/Admin

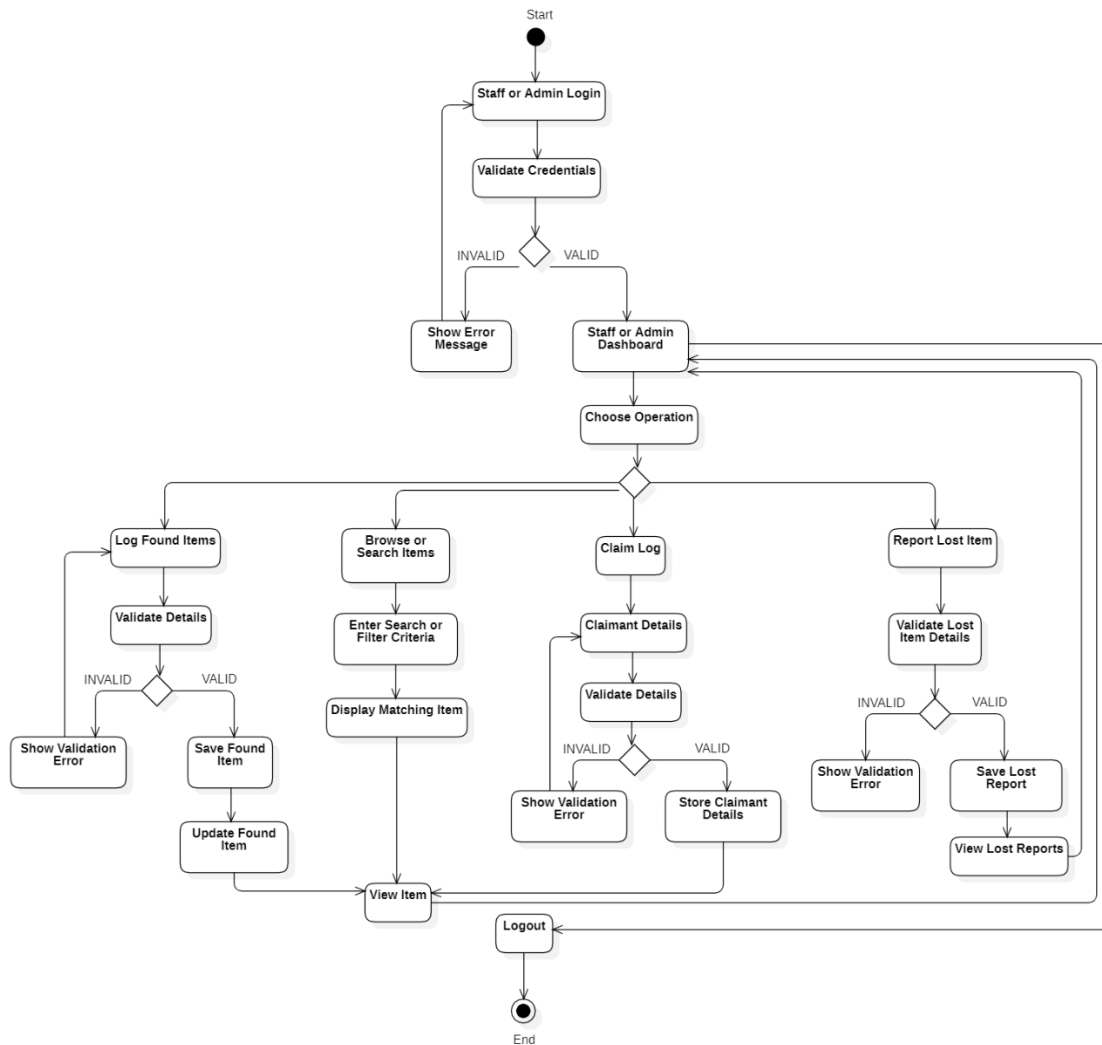


Figure 5.3.2.1 Activity Diagram - Admin / Staff

5.3.3 Admin

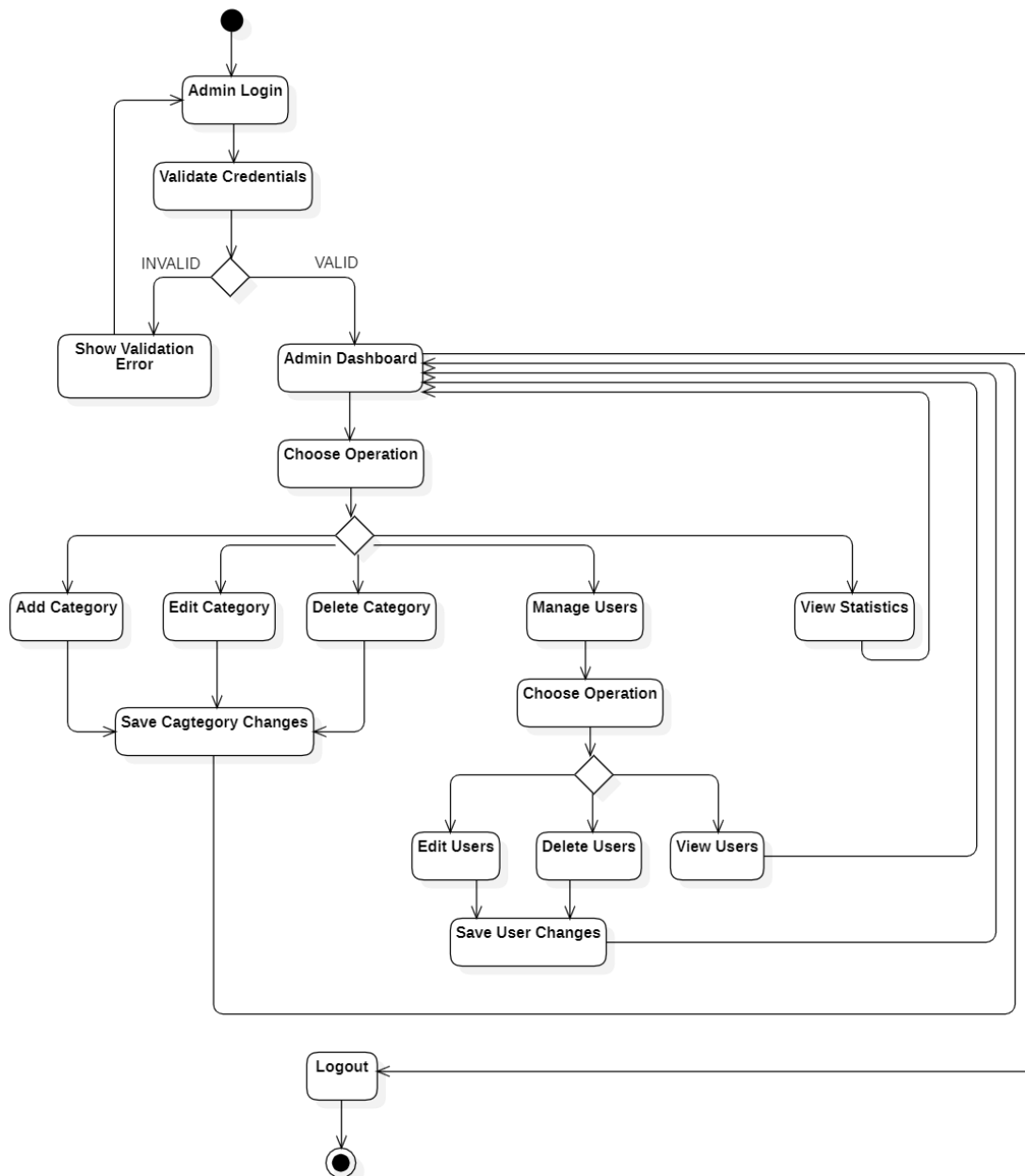


Figure 5.3.3.1 Activity Diagram - Admin

5.4 Sequence Diagram

5.4.1 Student / User

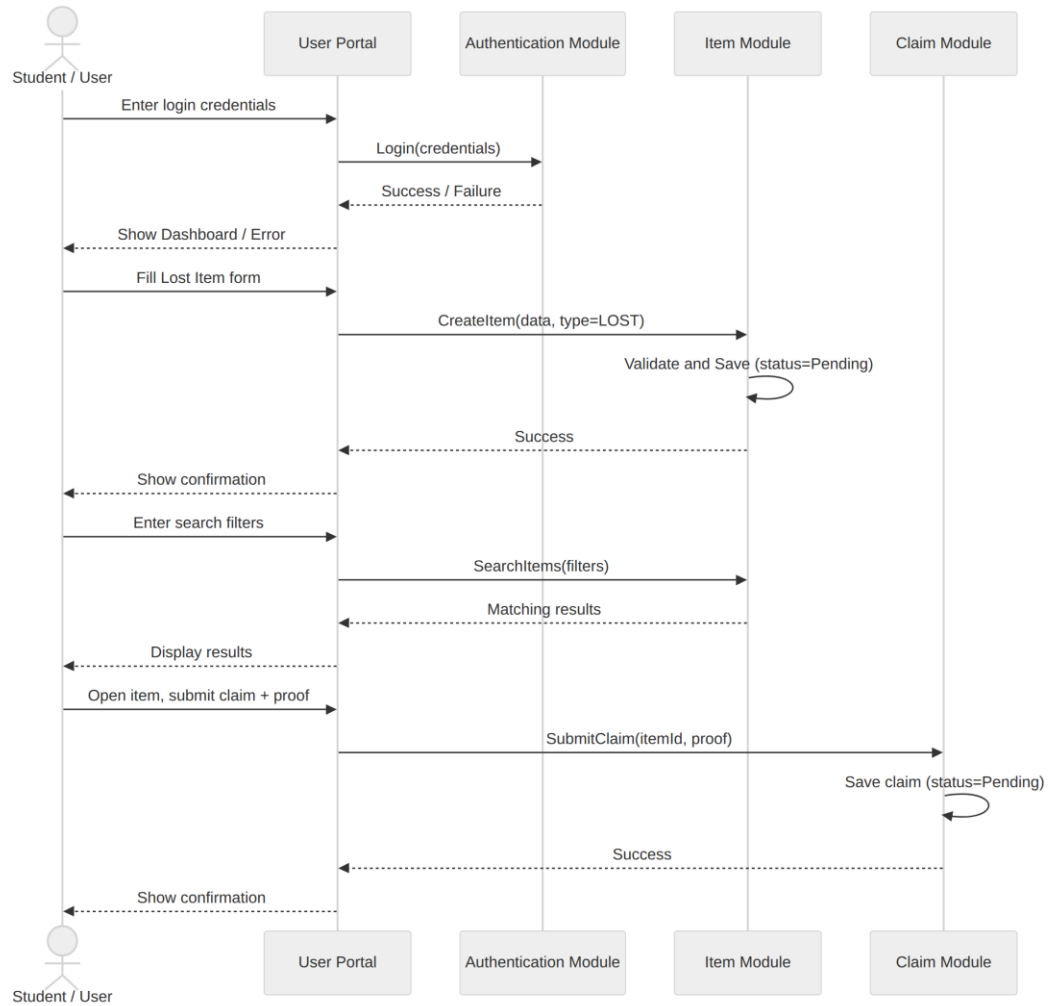


Figure 5.4.1.1

5.4.2 Admin / Staff

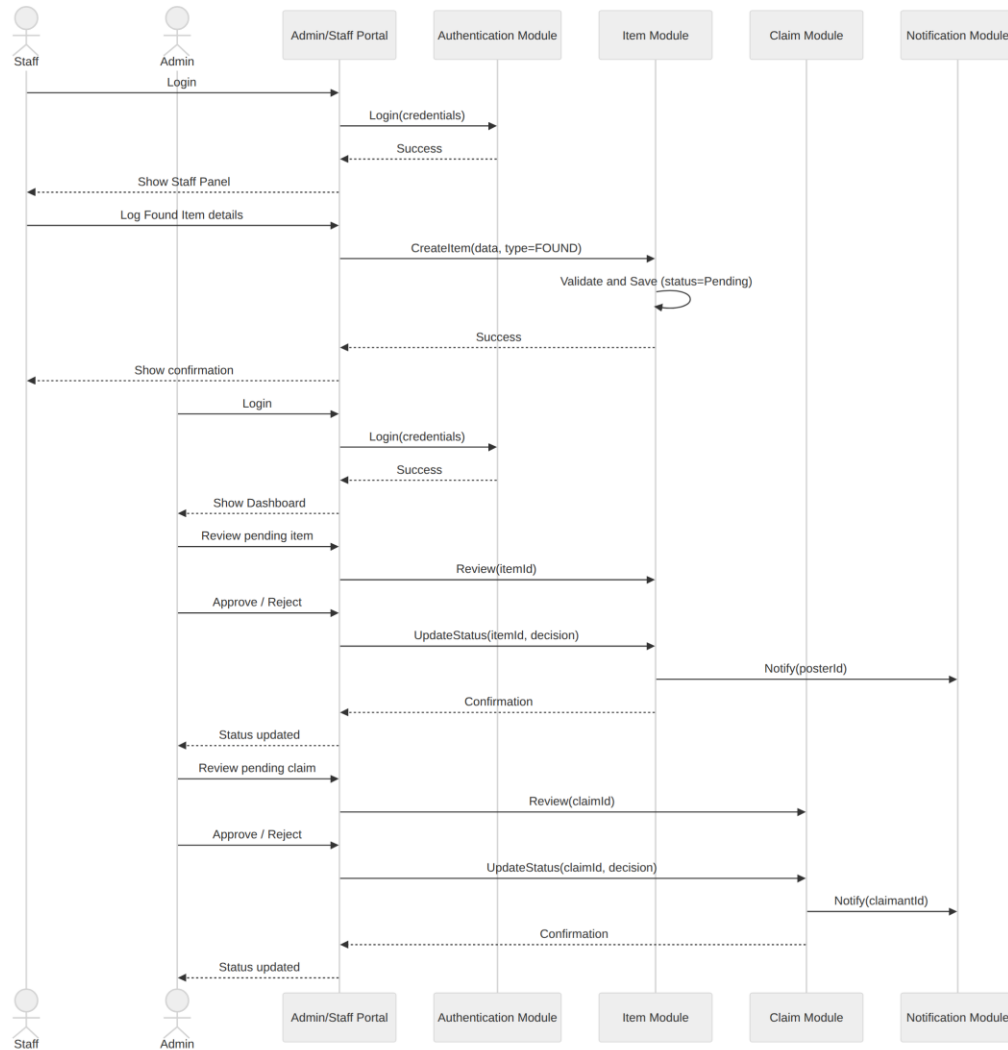


Figure 5.4.2.1 Sequence Diagram - Admin / Staff

5.5 ENTITY - RELATIONSHIP (E - R) DIAGRAM

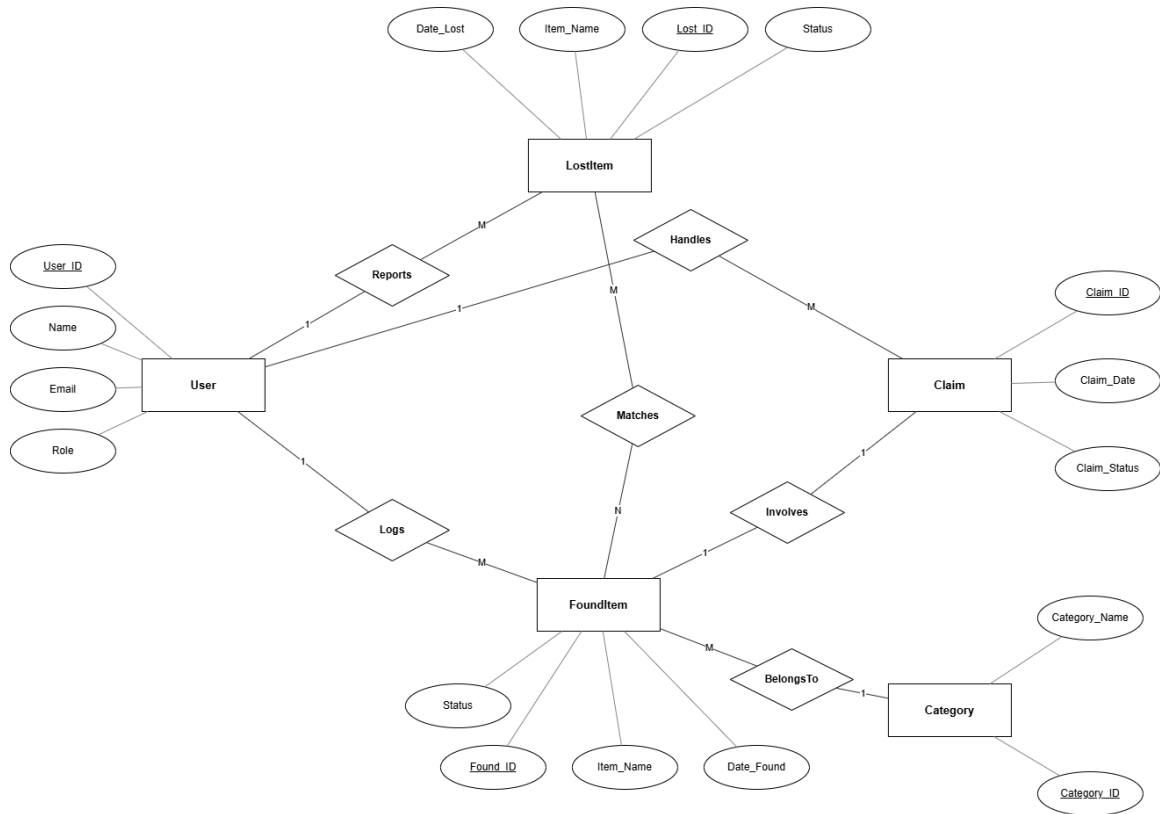


Figure 5.4.1 ENTITY-RELATIONSHIP DIAGRAM (E - R)

6. DATA DICTIONARY

This chapter defines the tentative database structure of FINDLY based on the functional modules, role permissions and item-management workflow described in Chapter 4. The database is designed around six main entities: **User, Item, Category, Handover, Notification and Audit Log**.

The schema supports the three system roles — **Student, Staff and Admin** — while maintaining the distinction between Lost-item reporting and Found-item logging. It also stores the custody and handover information associated with Found items without introducing a separate claim-submission workflow.

6.1 TABLE: USER

The USER table stores the account information of all registered users. The role field determines the features and operations available to the account.

Column	Data Type	Constraints	Description
userId	INT	PK, AUTO_INCREMENT	Unique identifier for a user account
name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(150)	NOT NULL, UNIQUE	Institute email used as login ID
password	VARCHAR(255)	NOT NULL	Salted password hash (e.g., bcrypt)
role	ENUM('STUDENT','STAFF','ADMIN')	NOT NULL, DEFAULT 'STUDENT'	Determines access level and permitted system functions
contactNo	VARCHAR(15)	NULLABLE	Contact phone number
accountStatus	ENUM('ACTIVE','INACTIVE','SUSPENDED')	NOT NULL, DEFAULT 'ACTIVE'	Current account access status
createdAt	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

Table 6.1 User Data Dictionary

6.2 TABLE: ITEM

The ITEM table is the **central table of FINDLY**. It stores both Lost and Found item records. The same table is used for both types of postings, with itemType distinguishing between LOST and FOUND.

For Lost items, postedBy identifies the user who reported the item. For Found items, loggedByStaff identifies the Staff/Admin account that logged the item.

Recipient information is also stored in this table because FINDLY does not maintain a separate Claim entity. These fields are populated only when a Found item reaches the handover stage.

Column	Data Type	Constraints	Description
itemId	INT	PK, AUTO_INCREMENT	Unique identifier for an item record
title	VARCHAR(150)	NOT NULL	Short title of the lost/found item
description	TEXT	NOT NULL	Detailed description of the item
itemType	ENUM('LOST','FOUND')	NOT NULL	Identifies whether the record is a Lost or Found item
categoryId	INT	FK → CATEGORY.categoryId, NOT NULL	Category to which the item belongs
location	VARCHAR(200)	NOT NULL	Location where the item was lost or found
itemDate	DATE	NOT NULL	Date the item was lost or found
imageUrl	VARCHAR(255)	NULLABLE	Path to the optional uploaded item image
custodyLocation	VARCHAR(200)	NULLABLE; required for FOUND items	Current location where a Found item is held
custodyStatus	ENUM('IN_CUSTODY','MOVED','HANDOVER')	NULLABLE; applicable to FOUND items	Current custody state of a Found item
status	ENUM('PENDING','ACTIVE','CLAIMED','RESOLVED','REJECTED')	NOT NULL, DEFAULT 'PENDING'	Current moderation and lifecycle state

Column	Data Type	Constraints	Description
postedBy	INT	FK → USER.userId, NOT NULL	User who reported a LOST item; may be Student, Staff or Admin
loggedByStaff	INT	FK → USER.userId, NULLABLE	Staff/Admin who logged a FOUND item
recipientName	VARCHAR(100)	NULLABLE	Name recorded for the recipient during handover
recipientContactNo	VARCHAR(15)	NULLABLE	Contact number recorded for the recipient
recipientEnrollmentNo	VARCHAR(30)	NULLABLE	Enrollment number recorded for the recipient
claimedAt	DATETIME	NULLABLE	Timestamp when the item was marked as Claimed
resolvedAt	DATETIME	NULLABLE	Timestamp when the item was marked as Resolved
createdAt	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Record creation timestamp
updatedAt	DATETIME	NULLABLE	Timestamp of the latest update

[Table 6.2 Item Data Dictionary]

6.3 TABLE: CATEGORY

The CATEGORY table stores the categories used for item classification. Administrators can add, edit and delete categories, while Students and Staff use the available categories when reporting or searching for items.

Column	Data Type	Constraints	Description
categoryId	INT	PK, AUTO INCREMENT	Unique identifier for a category

categoryName	VARCHAR(80)	NOT NULL, UNIQUE	Category name used in item posting and filtering
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Table 6.3 Category Data Dictionary

6.4 TABLE: NOTIFICATION

The NOTIFICATION table stores in-app notifications generated by the system. Notifications are associated with users and, where applicable, with the item that caused the notification. Examples include notifications related to item status changes, moderation decisions and other relevant system updates.

Column	Data Type	Constraints	Description
notificationId	INT	PK, AUTO_INCREMENT	Unique identifier for a notification
userId	INT	FK → USER.userId, NOT NULL	User who receives the notification
itemId	INT	FK → ITEM.itemId, NULLABLE	Item associated with the notification, where applicable
message	VARCHAR(255)	NOT NULL	Notification text shown to the user
isRead	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether the notification has been read
createdAt	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Time when the notification was generated

Table 6.3 Notification Data Dictionary

6.5 AUDIT LOG

The AUDIT_LOG table provides an accountability record for important actions performed within FINDLY.

Since the system requires actions such as posting, moderation, custody updates, claiming and resolution to be traceable, the audit log records the responsible user, related item, action performed and timestamp.

This table is particularly important for Staff/Admin operations involving Found items and handovers.

Column	Data Type	Constraints	Description
auditId	INT	PK, AUTO_INCREMENT	Unique identifier for an audit entry
userId	INT	FK → USER.userId, NOT NULL	User account that performed the action
itemId	INT	FK → ITEM.itemId, NULLABLE	Related item, where applicable
action	VARCHAR(100)	NOT NULL	Action performed, e.g., POST, UPDATE_CUSTODY, MODERATE, CLAIM, RESOLVE
oldStatus	VARCHAR(30)	NULLABLE	Previous item status, where applicable

Column	Data Type	Constraints	Description
newStatus	VARCHAR(30)	NULLABLE	New item status, where applicable
details	TEXT	NULLABLE	Additional information about the action
timestamp	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Date and time when the action was recorded

Table 6.3 Audit_Log Data Dictionary

7. PROTOTYPE - FIGMA

8. IMPLEMENTATION (USER MANUAL)

This chapter describes the implementation status of FINDLY and provides the basic environment setup and operating procedure required to run the system locally. The implementation follows the functional modules and role-based permissions defined in Chapter 4. The Student, Staff and Admin interfaces provide only the operations permitted to their respective roles.

8.1 IMPLEMENTATION STATUS

The implementation of FINDLY is being carried out progressively alongside the development and documentation of the project. The current implementation status is as follows:

- The MySQL database structure has been designed according to the entities and relationships defined in Chapter 6.
- The project folder structure has been organised using an MVC-inspired architecture, separating configuration, authentication, item management, administrative functions and supporting assets.
- The Registration and Login modules have been implemented with database connectivity, password hashing and role-based session handling.
- The Student Lost Item reporting module has been implemented with form validation and database insertion.
- The Staff/Admin Found Item module is being developed to allow authorised Staff and Admin users to log Found items along with custody information.
- The Search and Browse module is being developed to allow users to search and filter active Lost and Found listings.
- The Staff/Admin custody workflow is being implemented through custody updates, item claiming, recipient-detail recording and final resolution of Found items.
- The Admin moderation module is being developed for approving or rejecting pending item records.
- The Category Management and User Management modules are being implemented for Admin users.
- The Staff limited dashboard and Admin dashboard/statistics are being developed according to the role-based permission matrix.
- The Notification module is being integrated to provide users with updates regarding relevant item and account activities.
- The Figma-designed interfaces documented in Chapter 7 are being converted into the corresponding HTML, CSS, Bootstrap and JavaScript-based web pages.

The remaining implementation work will be completed progressively and integrated with the database and role-based access-control system.

8.2 DEVELOPMENT ENVIRONMENT

Component	Technology / Tool
Operating Environment	Windows / compatible operating system
Web Server	Apache
Server-Side Language	PHP
Database	MySQL
Development Package	XAMPP
Database Administration	phpMyAdmin
Front End	HTML5, CSS3, Bootstrap 5
Client-Side Scripting	JavaScript / jQuery
Code Editor	VS Code / compatible editor
Version Control	Git & GitHub
UI/UX Design	Figma
API / Endpoint Testing	Postman

Table 8.2.1 Development Environment

8.3 ENVIRONMENT SETUP

The following steps are required to configure FINDLY for local execution.

Step 1 - Install XAMPP

Install XAMPP containing:

- Apache
- MySQL
- PHP
- phpMyAdmin

After installation, open the XAMPP Control Panel.

Step 2 - Start Required Services

Start the following services:

- Apache
- MySQL

Both services must be running before accessing FINDLY.

Step 3 - Place the Project in htdocs

Copy the FINDLY project folder into the XAMPP web directory:

- C:\xampp\htdocs\findly\

The final project structure should follow the project's implemented folder organisation.

Step 4 - Create the Database

Open phpMyAdmin through the local XAMPP installation and create the FINDLY database.

- Import the provided SQL database file:
- findly.sql

The database should contain the tables defined in Chapter 6.

Step 5 - Configure Database Connection

Open the project's database configuration file:

- /config/db.php

Configure the local database connection parameters, including:

- Database host
- Database username
- Database password
- Database name

Step 6 - Start the Application

Once Apache and MySQL are running and the database has been configured, open a web browser and navigate to:

- <http://localhost/findly/>

The FINDLY authentication interface will be displayed.

8.4 User Manual

8.4.1 Student/User

A Student/User can perform the following operations:

- Register an account.
- Login to the system.
- Report a Lost Item.
- Browse Lost and Found listings.
- Search and filter items.
- View item details.
- Edit their own Lost Item report.
- Delete their own Lost Item report.
- View their profile and posting history.
- View notifications.
- Logout.
- The Student does not have access to:
- Log Found Item
- Update Found Item/Custody
- Mark Item as Claimed
- Record Recipient Details
- Mark Item as Resolved
- Manage Categories
- Manage Users
- Admin dashboard

8.4.2 *Staff/ Peon*

A Staff/Peon account has all general browsing and Lost Item reporting capabilities available to the Student, along with Found-item custody operations.

Staff can:

- Login to the system.
- Report a Lost Item.
- Log a Found Item.
- View Found Items.
- View Lost Reports.
- View Item Details.
- Search and filter listings.
- Update Found Item/Custody information.
- Mark a Found Item as Claimed.
- Record Recipient Name.
- Record Recipient Contact Number.
- Record Recipient Enrollment Number.

- Mark an item as Resolved.
- View limited Dashboard Statistics.
- Logout.
- Staff cannot:
- Manage Categories
- Manage Users
- Access the full Admin dashboard.

8.4.3 Admin

The Admin has the complete system-level access defined in the role matrix. In addition to the common Lost/Found and handover operations, the Admin can:

- View the Admin Dashboard.
- Moderate pending item posts.
- Approve or reject pending items.
- Add categories.
- Edit categories.
- Delete categories.
- View users.
- Activate Staff accounts.
- Suspend/reactivate users.
- View institution-wide statistics and reports.

The Admin can also perform the Found-item and handover activities available to Staff.

8.5 Role-Based Access

FINDLY applies role-based access control at the server side. The interface displays only the functions applicable to the logged-in role; however, access restrictions are also checked on the server to prevent unauthorised direct requests.

The three principal roles are:

- Student
- Staff / Peon
- Admin

The detailed permissions for these roles are defined in the Role-Based Permission Matrix presented in Chapter 4.

8.6 Basic Execution Flow

- The general execution procedure for the system is:
- Start XAMPP
- Start Apache and MySQL
- Open FINDLY in Browser
- Register / Login
- System Identifies User Role
- Role-Specific Interface
- Perform Permitted Operations
- Database Updated
- Display Updated Information / Notification
- Logout

9. SOFTWARE TESTING

This chapter presents the test cases prepared for verifying the functional and non-functional requirements of FINDLY. The test cases are derived directly from the requirements and role-based permission matrix specified in Chapter 4.

Testing focuses particularly on:

- Authentication
- Lost-item reporting
- Found-item logging
- Search and browsing
- Role-based access control
- Found-item custody management
- Handover and resolution
- Admin moderation
- Category and user management
- Dashboard statistics
- Security
- Usability
- Privacy
- File validation

The test cases will serve as the baseline for functional verification as the implementation progresses.

9.1 FUNCTIONAL TEST CASES

TC ID	Req.	Module	Scenario	Expected Result
TC01	1.1	Authentication	Register with valid Student details	Account is created successfully and user is redirected to Login.
TC02	1.1	Authentication	Register with duplicate or invalid institute email	Registration is rejected and a clear validation error is displayed.
TC03	1.2	Authentication	Login with valid active credentials	User is authenticated and redirected to the

TC ID	Req.	Module	Scenario	Expected Result
				appropriate role interface.
TC04	1.2	Authentication	Login with incorrect password or inactive account	Access is denied and an appropriate error message is displayed.
TC05	1.3	Authentication	User clicks Logout	Session is destroyed and the user is redirected to the Login page.
TC06	2.1	Student/User	Student reports a Lost Item with valid details	Lost Item is saved with itemType = LOST and status = PENDING.
TC07	2.1	Student/User	Student submits Lost Item with missing/invalid details	Submission is rejected and validation errors are displayed.
TC08	2.2	Search/Browse	User searches using keyword, category, type or location filters	Only matching active listings are displayed.
TC09	2.3	Student/User	Student edits their own Lost Item	Selected post is updated successfully and confirmation is displayed.
TC10	2.3	Access Control	Student attempts to edit another user's post	Request is denied and the other user's post remains unchanged.
TC11	2.4	Student/User	User opens My Profile/History	User's own posts and their current statuses are displayed.

TC ID	Req.	Module	Scenario	Expected Result
TC12	2.5	Notifications	User opens notification panel	Relevant notifications are displayed and opened notifications are marked as read.
TC13	3.1	Staff/Admin	Staff logs a Found Item with valid details	Found Item is saved with itemType = FOUND and status = PENDING.
TC14	3.1	Access Control	Student attempts to access Found-item logging directly	Request is denied server-side and no Found Item is created.
TC15	3.2	Staff/Admin	Staff updates custody information for a Found Item	Custody information is updated successfully.
TC16	3.3	Staff/Admin	Staff/Admin marks an Active Found Item as Claimed	Item status changes to CLAIMED and it is removed from active public listings.
TC17	3.4	Staff/Admin	Staff/Admin records recipient details for a Claimed Item	Recipient details are stored successfully and remain restricted to authorised users.
TC18	3.5	Staff/Admin	Staff/Admin resolves a Claimed Item after recipient	Item status changes to RESOLVED and the handover

TC ID	Req.	Module	Scenario	Expected Result
			details are recorded	timestamp is recorded.
TC19	3.6	Staff	Staff opens Dashboard Statistics	Only permitted Staff-level statistics are displayed.
TC20	4.1	Admin / staff Moderation	Admin approves a Pending Item	Item status changes to ACTIVE and becomes visible in public listings.
TC21	4.1	Admin Moderation	Admin rejects a Pending Item	Item status changes to REJECTED and the relevant user is notified.
TC22	4.2	Category Management	Admin adds, edits or deletes a category	Category changes are validated and reflected in item forms and search filters.
TC23	4.3	User Management	Admin activates, suspends or reactivates a user	Selected user's account status is updated and access changes accordingly.
TC24	4.4	Admin Dashboard	Admin opens Dashboard and Reports	Institution-wide statistics and relevant activity information are displayed correctly.

Table 9.1.1 Functional Test Cases

9.2 NON-FUNCTIONAL TEST CASES

TC ID	Req.	Category	Scenario	Expected Result
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TC25	NFR-01	Security	Inspect stored passwords in the database	Passwords are stored as salted hashes and never as plaintext.
TC26	NFR-02	Security	Submit malicious SQL/XSS-style input through a form	Input is safely validated/parameterised and no unintended query or script execution occurs.
TC27	NFR-03	Access Control	Student directly requests a Staff/Admin-only Found-item endpoint	Server-side role validation rejects the request.
TC28	NFR-04	Performance	Execute a normal filtered search under expected load	Search results are returned within the specified 2-second target under normal load.
TC29	NFR-05	Usability	Open the system on a 375px-wide mobile viewport	Layout remains responsive and features remain usable.
TC30	NFR-07	Session Security	Leave an authenticated session idle beyond the configured timeout	Session expires and the user must log in again.
TC31	NFR-09	File Upload	Upload an unsupported file type or image larger than 5 MB	Upload is rejected with a clear error message.
TC32	NFR-11	Privacy	Student views a Found Item containing recipient information	Recipient name, contact number and enrollment number are not exposed to Student/public users.

Table 9.2.1 Non-Functional Test Cases

9.3 Testing Approach

FINDLY will use the following testing approach:

- Functional Testing
- Each functional requirement will be tested using valid and invalid inputs to verify that the system produces the expected result.
- Role-Based Access Testing
- Each major operation will be tested against the three roles:
- Student

- Staff / Peon
- Admin
- This ensures that users cannot access operations outside their assigned permissions.
- Validation Testing

Forms will be tested with:

- Missing fields
- Invalid values
- Duplicate values
- Unsupported files
- Oversized images
- Security Testing

Security testing will verify:

- Password hashing
- SQL injection protection
- Input validation
- Session protection
- Server-side role verification
- Restricted recipient information
- Usability Testing

The interface will be tested on different screen sizes, particularly mobile screens, to ensure that the Bootstrap-based responsive layout remains usable.

9.4 Expected Testing Outcome

The objective of testing is to ensure that FINDLY:

- Correctly authenticates users.
- Applies the defined Student, Staff and Admin permissions.
- Allows Students to report Lost Items.
- Allows only Staff/Admin to log Found Items.
- Correctly manages Found-item custody.
- Correctly handles Claimed and Resolved statuses.
- Prevents unauthorised access to restricted functions.
- Allows Admins to moderate items and manage categories/users.
- Protects user and recipient information.
- Meets the specified security, usability and performance requirements.

10. CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

FINDLY – Campus Lost & Found is a proposed centralised web-based system for managing lost and found items within the campus. The system provides a structured platform for reporting Lost Items, logging Found Items, searching listings, managing item status and recording final handovers.

The system follows a role-based approach where Students can report Lost Items, while Staff/Peon and Admin can log Found Items and manage the subsequent item-handling process. Admin additionally manages post moderation, users, categories and overall system statistics.

The requirements, technological specifications, database design, UML diagrams, prototype screens and testing plan presented in this report provide the foundation for implementing the proposed system.

10.2 FUTURE WORK

- Email/SMS notification integration.
- Automated Lost–Found item matching.
- QR-code-based item identification.
- Dedicated Android/mobile application.
- Advanced Admin reports and analytics.

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